

# GAURAB ACHARYA

Vancouver, BC, Canada

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## EDUCATION

### University of British Columbia

Bachelor of Applied Science in Biomedical Engineering — Specialized in Systems and Signals

Vancouver, British Columbia

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL, C#, JavaScript, R, Swift

**Frameworks:** Spring Boot, JUnit, Model Context Protocol (MCP), React, TensorFlow

**Technologies:** Apache Airflow, AWS S3, Docker, Jenkins, PostgreSQL, Oracle, DuckDB

## EXPERIENCE

### QuoteMedia

Jun. 2025 - Present

Software Engineer

Vancouver, BC

- Owned batch **ETL** from **Oracle** to **AWS S3** using **Apache Airflow**, to create a file store of static data supporting MCP servers
- Built and operated Dockerized internal MCP servers using **FastMCP** and **n8n** to automate reporting, cutting manual effort by **80%**
- Developed a **Java Spring Boot API**, combining **DuckDB** and **S3** to cut query time from **40s to 10s**, accelerating analytics
- Validated reliability of API's with **JMeter** and built a custom connection-pooling service to handle concurrent database requests
- Implemented a data-quality (DQ) pipeline from source systems into **PostgreSQL**, to surface and triage issues earlier in the lifecycle

### QuoteMedia

May 2024 – Aug. 2024, May 2022 – Apr. 2023

Software Developer Intern

Vancouver, BC

- Developed a data pipeline using **SQL**, **Python**, and **Apache Airflow**, resulting in **20x** aggregation speed for analytic queries
- Built and maintained backend services & REST APIs (**Java**, **Spring Boot**, **SQL**) used in high-traffic market-data delivery
- Optimized query performance 10x by migrating legacy databases (**Oracle to PostgreSQL**) and tuning schema
- Created a performance testing pipeline (**Vegeta & Airflow**) and observability dashboards (**InfluxDB/Grafana**) to shorten feedback loops and improve deployment confidence
- Built **Kibana/Grafana** dashboards summarizing API speed and usage to inform capacity planning and SLA adherence

### Vantech Medical

Sep. 2023 – Apr. 2024

Augmented Reality Developer

- Designed an AR application using **Unity**, **C#**, **ARKit** rendering vein images, improving vein detection for intravenous procedures
- Integrated an Infrared (IR) Vein Detection Model on **Google Cloud Platform** achieving  $\leq 1s$  latency from capture to projection
- Conducted iterative user testing with healthcare providers to refine usability and ensure accuracy in clinical environments

### ProCogia

May 2023 – Sep. 2023

R Developer Intern

Vancouver, BC

- Developed an **R** package integrating **OpenAI ChatGPT API** to analyze and debug R scripts in **<5 seconds**, increasing efficiency in developing data driven projects
- Engineered system prompts and conversational memory to improve AI response consistency by **90+%** across diverse use cases
- Took full ownership of package architecture, testing, and deployment

## PROJECTS

**AI-Powered Medical Chatbot**  | *React, Next.js, Tailwind CSS, Python, LangChain, Flask, Pinecone, Google Gemini*

- Implemented a **RAG pipeline** to deliver medical information with source-linked responses, improving citation quality by **20%**
- Indexed the Gale Encyclopedia of Medicine using the vector database, **Pinecone**, and **Hugging Face embeddings**, to allow retrieval from a **Flask API** , which delivered medical information in real-time, with context-aware responses
- Designed **Figma wireframes**  and translated them into dynamic UI components with component-driven architecture
- Built a modern, responsive chat interface using **Next.js** and **Tailwind CSS**, to allow users to use the service easily

## LEADERSHIP / EXTRACURRICULAR

### Basketball / Coaching

Apr. 2016 – Present

*RBL Basketball* , *Drive Basketball* , *Strathcona Community Centre*, *Dreamers Basketball* 

- Coached over 2000 players in basketball camps and teams, working with four different clubs in Vancouver for over 9 years
- 4x British Columbia Provincial Basketball Championship All-Star